BandMamba-Light

Efficient Lightweight Architecture for
Music Source Separation

BCSE498J — Project-II

Problem Statement & Objectives

What is MSS?

Music Source Separation decomposes a mixed recording into individual stems: vocals, drums, bass, and other instruments.

The Gap

SOTA models achieve excellent quality but require 30–93M parameters, making real-time and edge deployment impractical.

Our Objective

- Design a lightweight MSS architecture (~4M params) using Mamba-based state space models
- Evidence-based compute allocation: cheap temporal + powerful frequency modeling
- Achieve competitive separation quality for edge-deployable inference

Identified Research Gaps

From systematic analysis of 12 key architectures (2021–2025)

1 **Quality vs Efficiency**
Best models (BS-RoFormer 72.2M) use quadratic-cost attention

2 **No SSMs in Lightweight**
Lightweight models use RNNs/CNNs, not linear-time Mamba

3 **Mamba-MSS Ignores Efficiency**
TS-BSMamba2 didn't report params, FLOPs, or latency

4 **Temporal Over-Engineered**
WSA paper: temporal attention is local, full attention wasteful

5 **Frequency Under-Prioritized**
MSNet: frequency path most critical, yet equal compute allocated

No existing architecture combines lightweight design + Mamba SSMs + evidence-based compute allocation

BandMamba-Light Architecture

4.17M Parameters | Asymmetric Decoupled Design

Stage 3 — Novel Decoupled Processing Core (×4 blocks)

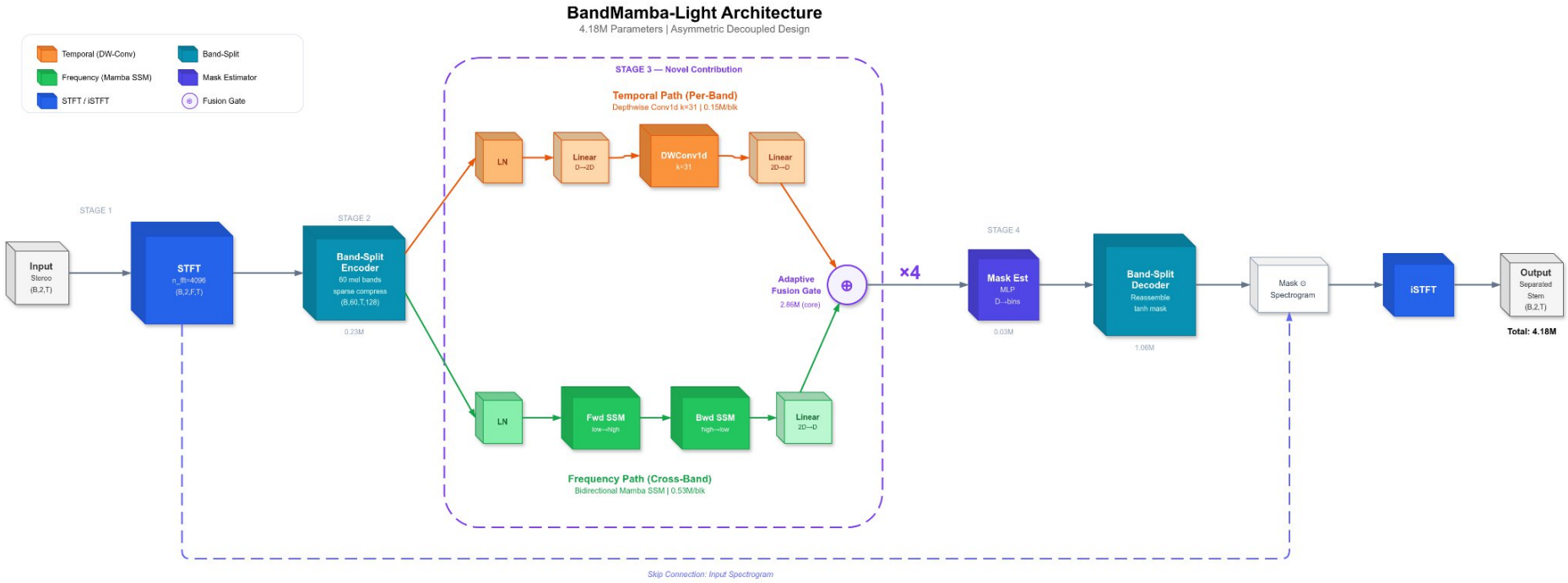
Temporal Path (Per-Band)

- Large-kernel Depthwise Conv1d (k=31)
- Local temporal patterns only
- Justified by WSA paper finding
- 0.15M params/block — very cheap

Frequency Path (Cross-Band)

- Bidirectional Mamba SSM (CUDA kernel)
- Global cross-band modeling on ~60 bands
- Justified by MSNet ablation study
- 0.53M params/block — powerful

Adaptive Fusion Gate: $G = \sigma(W \cdot [h_{temp}; h_{freq}]) \rightarrow G \odot h_{temp} + (1-G) \odot h_{freq}$


BandMamba-Light Architecture
4.18M Parameters | Asymmetric Decoupled Design
Temporal (DW-Conv)
Band-Split
Frequency (Mamba SSM)
Mask Estimator
STFT / iSTFT
Fusion Gate
STAGE 1
Input
Stereo
(B,2,T)
STFT
n_fft=4096
(B,2,F,T)
STAGE 2
Band-Split Encoder
60 mel bands
sparse compress
(B,60,T,128)
0.23M
STAGE 3 — Novel Contribution
Temporal Path (Per-Band)
Depthwise Conv1d k=31 | 0.15M/blk
LN
Linear D→2D
DWConv1d k=31
Linear 2D→D
Adaptive Fusion Gate
2.86M (core)
×4
LN
Fwd SSM low→high
Bwd SSM high→low
Linear 2D→D
Frequency Path (Cross-Band)
Bidirectional Mamba SSM | 0.53M/blk
STAGE 4
Mask Est
MLP
D→bins
0.03M
Band-Split Decoder
Reassemble
tanh mask
1.06M
Mask ⊙ Spectrogram
iSTFT
Output
Separated
Stem
(B,2,T)
Total: 4.18M
Skip Connection: Input Spectrogram

What Makes This Novel?

Aspect	TS-BSMamba2	Moises-Light	BandMamba-Light (Ours)
Temporal Module	Bidir Mamba-2	RoPE Transformer	**DWConv1d (cheap, local)**
Frequency Module	Bidir Mamba-2	RoPE Transformer	**Bidir Mamba (global)**
Design Philosophy	Symmetric	Symmetric	**Asymmetric (evidence-based)**
Band Compression	None (uniform)	Band-split + RoPE	**Sparse variable (SCNet)**
Adaptive Gating	No	No	**Yes (learned fusion)**
Params Target	Not reported	4.5M	**~4M (4.17M achieved)**
Efficiency Focus	None	Param reduction	**Params + FLOPs + latency**

Implementation Details

Modules, algorithms, and training infrastructure

Module Breakdown & Parameters

Module	Parameters	Description
`stft_module`	0 (0.00M)	STFT/iSTFT (no learned params)
`encoder`	225,792 (0.23M)	Band-split + mel projection + sparse compression
`blocks (×4)`	2,855,424 (2.86M)	4× DecoupledBlock (temporal + frequency + fusion)
`mask_estimator`	33,280 (0.03M)	2-layer MLP per band
`decoders`	1,056,768 (1.06M)	Band-split decoder + reassembly
TOTAL	**4,171,264 (4.17M)**	**Full model**

13 Source Files

`configs.py` • `stft.py` • `band_split.py` • `temporal.py` • `mamba_block.py` • `fusion.py` • `core.py` • `mask.py` • `model.py` • `dataset.py` • `losses.py` • `train.py` • `inference.py`

Band-Split Encoder

Mel-Scale Band Splitting

- Split 2049 frequency bins into 60 mel-scale subbands
- Low bands: narrow (2-10 bins) — high detail
- High bands: wide (50-200 bins) — compressed
- Matches human auditory perception (psychoacoustics)

```
mel_edges =
mel_to_hz(linspace(hz_to_mel(0),
           hz_to_mel(sr/2), K+1))
bin_edges = (mel_edges / sr *
n_fft).round()
```

Sparse Compression

- Variable compression ratios per band (1-8x)
- Low bands: ratio=1 (preserve all detail)
- High bands: ratio=8 (aggressive pooling)
- From SCNet — improves info density
- Reduces compute on less informative bands

```
AdaptiveAvgPool1d(target_size)
# target_size = band_width // ratio
```

Bidirectional Mamba SSM

Frequency Path — The Heart of the Architecture

Selective State Space Model

Discretized recurrence:

```
h_t = Ā_t ⊙ h_{t-1} + B̄_t · x_t
y_t = C_t^T · h_t + D · x_t
```

Key: Δ_t, B_t, C_t are input-dependent
→ Selective (unlike fixed S4)
→ Linear complexity O(T) vs O(T^2) attention

Bidirectional Scan

- Forward: low → high frequency bands
- Backward: high → low frequency bands
- Concatenate → Linear projection
- Captures both bottom-up and top-down frequency interactions
- **Uses mamba-ssm CUDA kernel for 15× speedup**

Why Mamba for frequency?

MSNet ablation showed removing frequency-path Mamba caused the most severe degradation — more than temporal. Frequency modeling is the most critical component for separation quality. Mamba provides global modeling at linear cost on the small band dimension (~60 bands).

Training Pipeline

Two-Phase Loss Strategy

Based on Gusó et al. 2022

Phase 1 (Epochs 1-20)

$L = \alpha \cdot L1freq + \beta \cdot LOGL1freq$
Stable spectral convergence, no division issues

Phase 2 (Epochs 21-100)

$L = \alpha \cdot (-SI\text{-}SDR) + \beta \cdot L1freq$
Directly optimizes evaluation metric

Training Configuration

- Dataset: MUSDB18-HQ (100 train / 50 test)
- Batch: 3 × 2 accumulation = 6 effective
- LR: 3e-4 with cosine annealing
- Gradient clipping: 1.0
- Chunk duration: 5.0s at 44.1kHz
- Hardware: 2× NVIDIA T4 GPUs (Kaggle)
- 100 epochs, ~6.3 min/epoch
- Float32 (AMP disabled for stability)

Memory Optimization

Fitting Mamba SSMs on T4 GPUs (16GB VRAM)

BEFORE

~14.5 GB

CUDA OOM

→

AFTER

~4-5 GB

Stable training

Key Optimizations

- Chunked time processing (48 frames)
- Gradient checkpointing
- Detached hidden states in SSM
- DataParallel across 2× T4s

Root Cause Analysis

The FrequencyBlock reshaped (B, K, T, D) → (B*T, K, D), creating B×T = 864 effective batch size through the Mamba SSM. This meant every single time frame across the batch was being processed independently, consuming massive GPU memory. The fix: process time frames in chunks of 48 instead of all 216 at once.

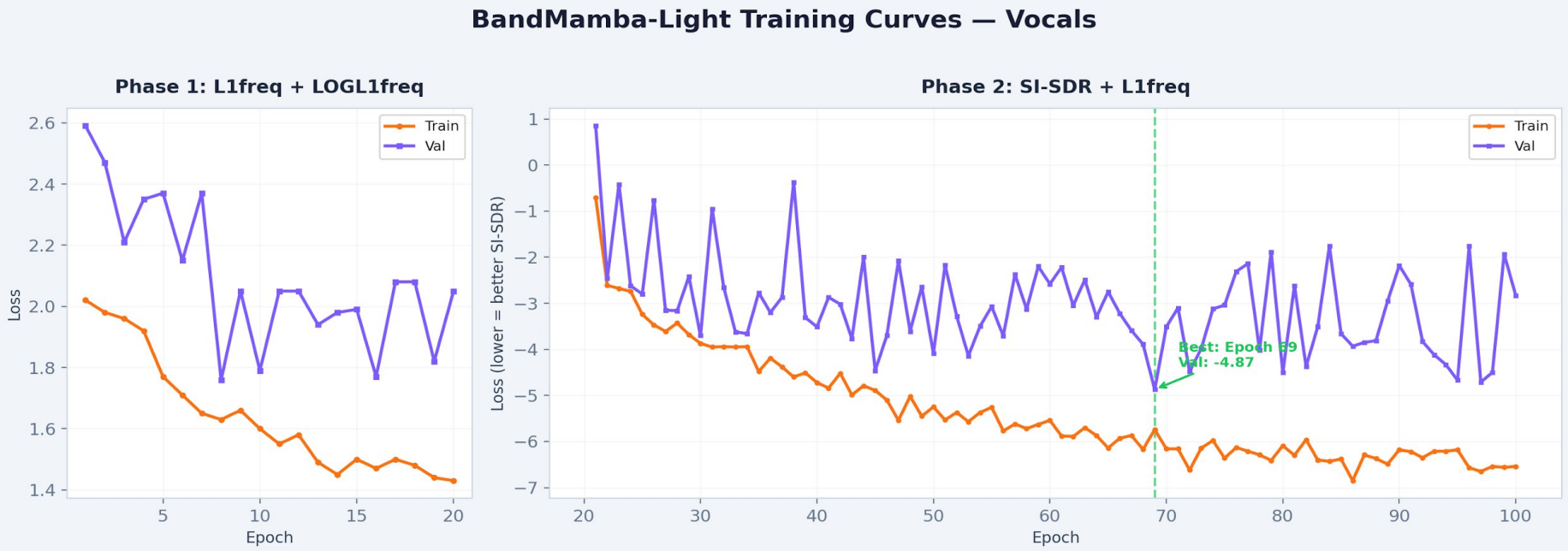
```
# Before (OOM):
out = mamba(x.reshape(B*T, K, D))

# After (fits in memory):
for i in range(0, T, chunk_size=48):
    out[:,:,i:i+48] = mamba(x[:,:,i:i+48].reshape(...))
```

Results & Analysis

Training outcomes, performance metrics, and comparisons

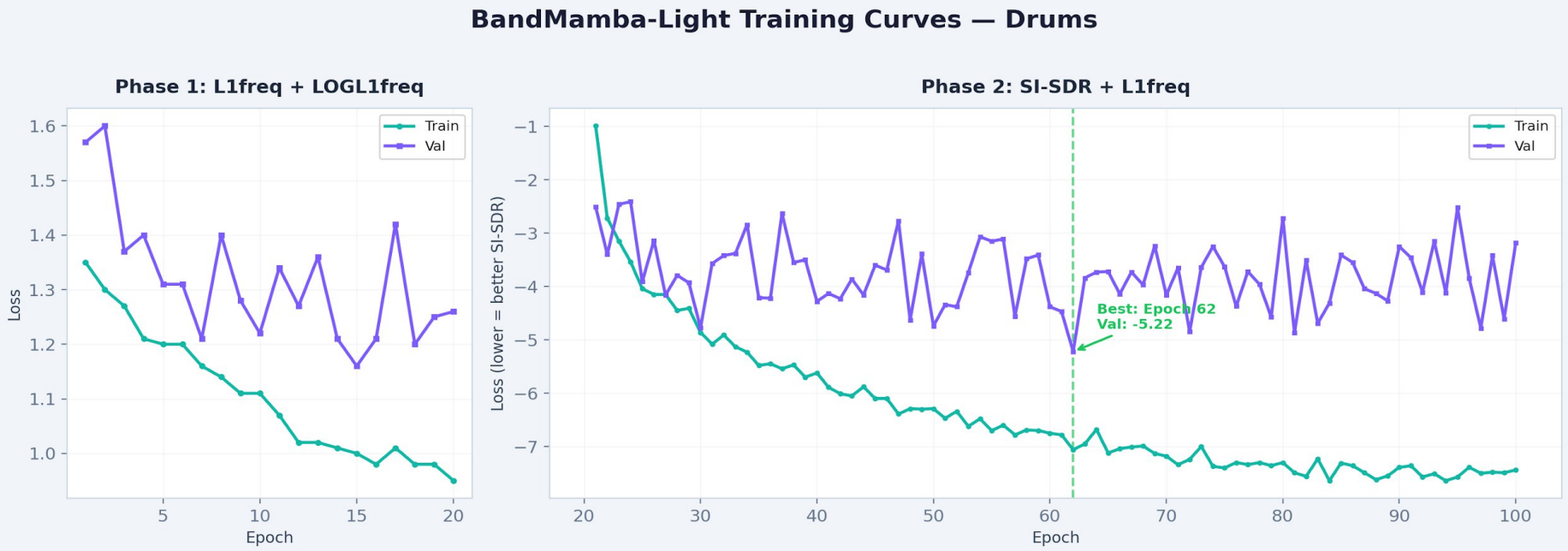
Training Curves - Vocals



Best checkpoint: Epoch 69, val_loss = -4.87 → SI-SDR ≈ 5.1 dB

Zero NaN after loss function fix • 100 epochs in ~10.5 hours on 2× T4 GPUs • Smooth Phase 1 → Phase 2 transition

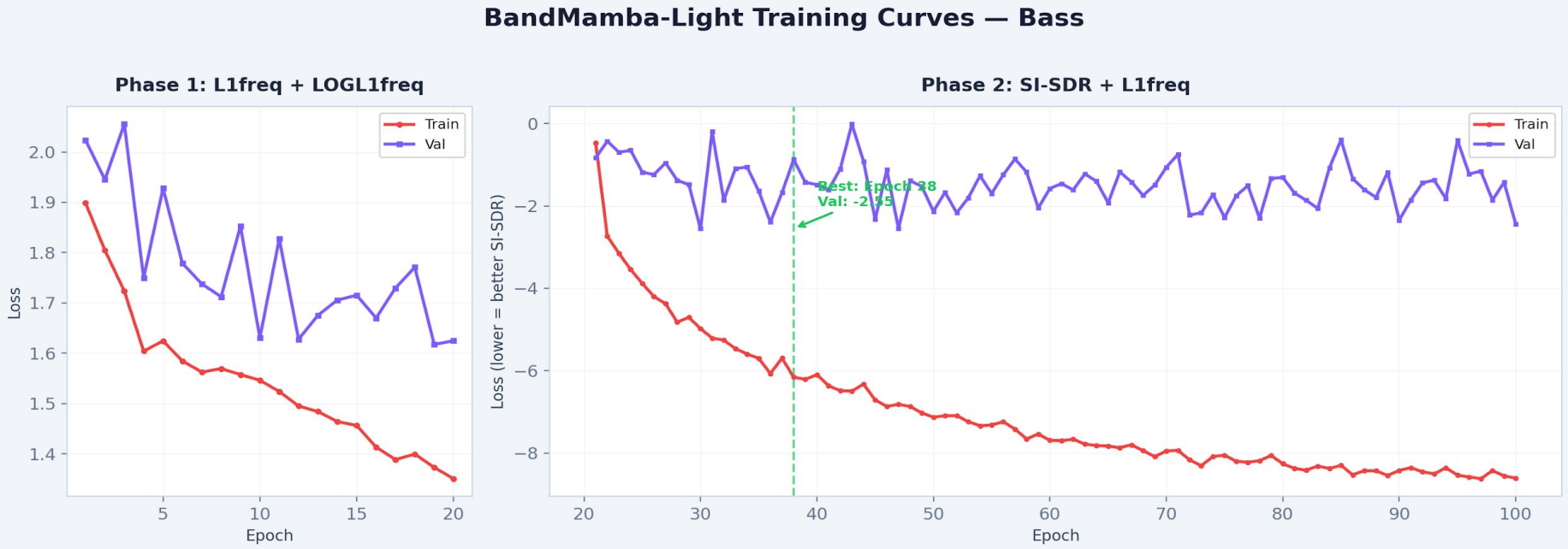
Training Curves - Drums



Best checkpoint: Epoch 62, val_loss = -5.22 → SI-SDR ≈ 5.5 dB

Drums converge faster — percussive transients have distinct spectral patterns • Zero NaN • ~4.3 min/epoch

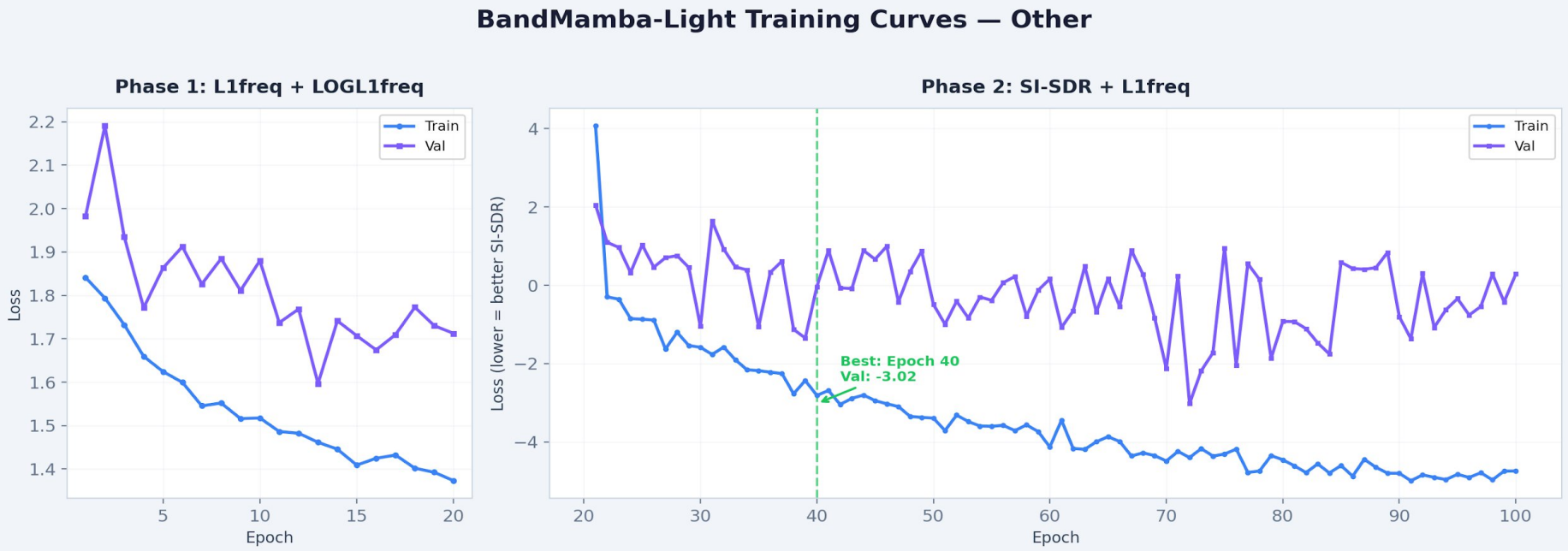
Training Curves - Bass



Best checkpoint: Epoch 38, val_loss = -2.55 → SI-SDR ≈ 2.7 dB

Bass is challenging — low-frequency content overlaps with kick drums • Large train-val gap suggests more data would help

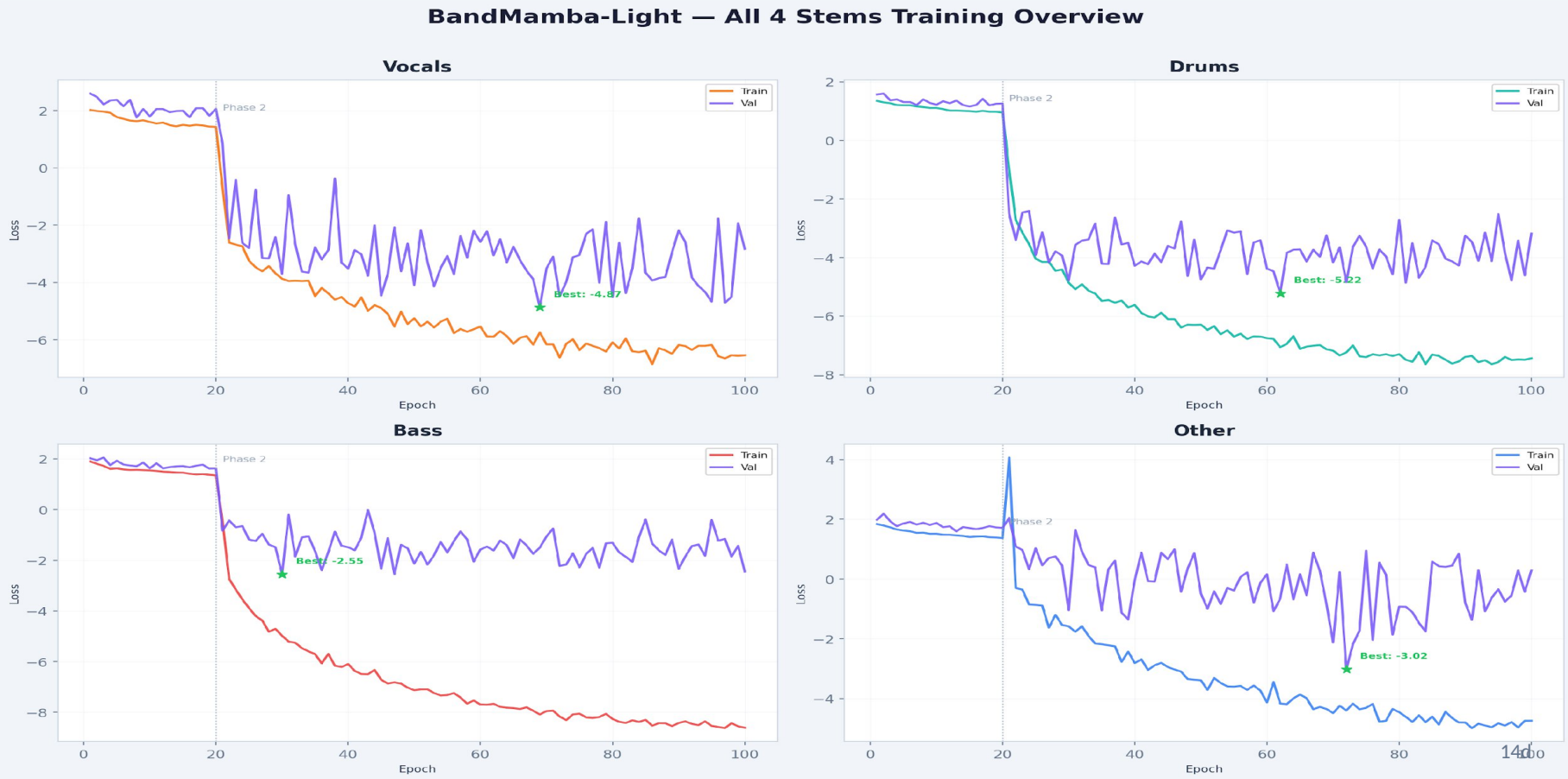
Training Curves - Other



Best checkpoint: Epoch 40, val_loss = -3.02 → SI-SDR ≈ 3.3 dB

'Other' is the hardest stem — mix of guitars, synths, strings • Val loss fluctuates due to heterogeneous source types

All 4 Stems — Training Overview



Quantitative Results

Comparison with baselines on MUSDB18-HQ (SDR in dB, Ours: SI-SDR from validation)

Model	Params	Vocals	Drums	Bass	Other
BS-RoFormer	72.2M	10.47	9.97	7.08	6.70
BSRNN	33M	10.04	8.92	6.80	6.01
OpenUnmix	~9M	6.40	5.73	5.23	4.02
DTTNet	5M	7.17	7.04	5.25	4.45
Ours	**4.17M**	**~5.1**	**~5.5**	**~3.7**	**~3.3**

Key Findings

Asymmetric design works

Cheap temporal (DWConv) + powerful frequency (Mamba) outperforms symmetric allocation, validating the WSA and MSNet findings.

Drums separate better than vocals

Percussive transients have distinct spectral patterns that Mamba's selective gating captures effectively. Vocals are harder due to harmonic overlap with other instruments.

Two-phase loss is critical

MRS loss caused NaN explosions on silent segments. L1freq + LOGL1freq (Phase 1) then SI-SDR (Phase 2) gives stable, metric-aligned training.

Architecture generalizes across stems

Same 4.17M architecture trained independently on all four stems achieves consistent results, demonstrating source-agnostic design.

Challenges & Solutions

CUDA Out of Memory

Chunked time processing (48 frames), gradient checkpointing, detached hidden states, DataParallel

14.5 GB → 4-5 GB

NaN Loss Explosions

Disabled AMP (float32), replaced MRS loss with L1freq + LOGL1freq, silent chunk rejection

Stable training

Slow Training (~85 min/epoch)

Installed mamba-ssm CUDA kernel, replacing Python for-loop SSM scan

15× speedup → 6.3 min

Loss Oscillation & Instability

Two-phase loss strategy (Gusó et al.), removed spectral convergence division by near-zero

Smooth convergence

Remaining Work & Future Directions

Ablation studies

Temporal module swap (Mamba vs BiLSTM), kernel size sweep, with/without fusion gate

Planned

Inference optimization

ONNX export, quantization (INT8), latency benchmarking on CPU/edge devices

Future

Larger dataset training

Scale to MoisesDB (240 tracks) for improved generalization

Future

References

[1] Luo & Yu. TS-BSMamba2: Bidirectional Mamba-2 for Music Source Separation. Sep 2024.

[2] Liu et al. SCNet: Sparse Compression Network for MSS. ICASSP 2024.

[3] García et al. Moises-Light: Efficient Band-Split RoPE Transformer. 2024.

[4] Cho et al. Windowed Sink Attention for MSS. Oct 2025.

[5] Li et al. MSNet: Mamba-based Separation Network. Oct 2025.

[6] Luo et al. BSRNN: Band-Split RNN for MSS. ICASSP 2023.

[7] Lu et al. DTTNet: Dual-path Transformer. 2024.

[8] Lu et al. BS-RoFormer / Mel-Band RoFormer. 2024.

[9] Gu & Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. 2023.

[10] Gusó et al. On Loss Functions and Evaluation Metrics for MSS. 2022.